

# RAGHAV ENGINEERING

Welding Procedure Specification (WPS) — In Accordance with ASME Section IX

|                   |                |                |            |
|-------------------|----------------|----------------|------------|
| Basic Information |                |                |            |
| WPS No.           | WPS/RE/301-E2E | PQR No.        | PQR/RE/301 |
| Welding Process   | SMAW           | Type           | Manual     |
| Revision          | Rev 0          | Effective Date |            |
| Date of Welding   |                |                |            |

|                 |                 |              |         |
|-----------------|-----------------|--------------|---------|
| Joints (QW-402) |                 |              |         |
| Groove Type     | Double V Groove | Root Gap     | 1.5-2mm |
| Root Face       |                 | Groove Angle | 60°±5°  |
| Groove Length   |                 | Groove Width |         |
| Backing         |                 | Retainer     |         |

|                      |           |                  |     |
|----------------------|-----------|------------------|-----|
| Base Metals (QW-403) |           |                  |     |
| Material Spec.       | ASTM A217 | Type or Grade    | WC6 |
| P.No.                |           | Heat No.         |     |
| Test Coupon Thk.     |           | Test Coupon Dia. |     |
| Base Material        | WC6       |                  |     |

|                        |        |                  |          |
|------------------------|--------|------------------|----------|
| Filler Metals (QW-404) |        |                  |          |
| SFA Spec.              | A5.5   | AWS Class.       | E8018-B2 |
| Filler F.No.           | 4      | Weld Metal A.No. | 3        |
| Filler Size            | 4.00mm | Feed Rate        |          |
| Weld Metal Thk.        |        | Filler Material  |          |

|                    |      |                  |  |
|--------------------|------|------------------|--|
| Position (QW-405)  |      |                  |  |
| Position of Groove | Flat | Weld Progression |  |

|                  |        |                 |        |
|------------------|--------|-----------------|--------|
| Preheat (QW-406) |        |                 |        |
| Preheat Temp.    | 150 °C | Interpass Temp. | 300 °C |
| Others           |        |                 |        |

|                                   |     |               |              |
|-----------------------------------|-----|---------------|--------------|
| Post Weld Heat Treatment (QW-407) |     |               |              |
| PWHT Required                     | Yes | Temperature   | 705 – 715 °C |
| Time Range                        |     | Cooling       |              |
| Rate of Heating                   |     | Loading Temp. |              |
| Unloading Temp.                   |     |               |              |

|                 |  |               |  |
|-----------------|--|---------------|--|
| Gas (QW-408)    |  |               |  |
| Shielding Gas   |  | Trailing Gas  |  |
| Backing Gas     |  | % Composition |  |
| Flow Rate (lpm) |  |               |  |

|                                     |           |               |              |
|-------------------------------------|-----------|---------------|--------------|
| Electrical Characteristics (QW-409) |           |               |              |
| Current (AC/DC)                     | DC        | Polarity      | Reverse (EP) |
| Amps (Range)                        | 160-185 A | Volts (Range) |              |
| Tungsten Elec. Size                 |           | Travel Speed  |              |
| Heat Input                          |           |               |              |

|                      |  |  |  |
|----------------------|--|--|--|
| Weld Pass Parameters |  |  |  |
|----------------------|--|--|--|

| Weld Pass | Process | Filler Class. | Dia. | Current/Pol. | Amps    | Volts | Travel Speed | Heat Input |  |
|-----------|---------|---------------|------|--------------|---------|-------|--------------|------------|--|
| Root pass | SMAW    |               | 3.15 |              | 100-140 |       |              |            |  |

Technique (QW-410)

|                   |  |                    |  |
|-------------------|--|--------------------|--|
| Bead Type         |  | Oscillation        |  |
| Pass/Side         |  | Layer              |  |
| Electrode         |  | Contact Tube Dist. |  |
| Orifice/Gas Cup   |  | Interpass Cleaning |  |
| Back Gouging      |  | Electrode Spacing  |  |
| Change of Process |  | Peening            |  |
| Transfer Mode     |  | Torch Orifice Dia. |  |
| Filler Delivery   |  | Thermal Process    |  |

Approval

|                                               |             |
|-----------------------------------------------|-------------|
| R. Naganathan / CEO<br>Prepared & Approved By | Reviewed By |
|-----------------------------------------------|-------------|